

Wenhan LUO, Ph.D., IEEE Senior Member

CONTACT INFORMATION	Email: whluo.china@gmail.com Homepage: https://whluo.github.io/						
WORKING EXPERIENCE	<table><tr><td>The Hong Kong University of Science and Technology • Associate Professor Sun Yat-sen University • Associate Professor Tencent • Senior Researcher Amazon • Research Scientist Tencent • Senior Researcher </td><td>Hong Kong Jun, 2024 – present Shenzhen, China Jul, 2022 – Mar, 2024 Shenzhen, China Mar, 2020 – Jul, 2022 Palo Alto, CA, USA Jul, 2019 – Jan, 2020 Shenzhen, China Aug, 2016 – May, 2019</td></tr></table>	The Hong Kong University of Science and Technology • Associate Professor Sun Yat-sen University • Associate Professor Tencent • Senior Researcher Amazon • Research Scientist Tencent • Senior Researcher	Hong Kong Jun, 2024 – present Shenzhen, China Jul, 2022 – Mar, 2024 Shenzhen, China Mar, 2020 – Jul, 2022 Palo Alto, CA, USA Jul, 2019 – Jan, 2020 Shenzhen, China Aug, 2016 – May, 2019				
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EDUCATION	<table><tr><td>Imperial College London • Ph.D. in Computer Science </td><td>London, UK Oct, 2012 – Jun, 2016</td></tr><tr><td>Institute of Automation, Chinese Academy of Sciences • M.E. in Control Theory and Control Engineering </td><td>Beijing, China Sep, 2009 – Jul, 2012</td></tr><tr><td>Huazhong University of Science and Technology • B.E. in Automation </td><td>Wuhan, China Sep, 2005 – Jul, 2009</td></tr></table>	Imperial College London • Ph.D. in Computer Science	London, UK Oct, 2012 – Jun, 2016	Institute of Automation, Chinese Academy of Sciences • M.E. in Control Theory and Control Engineering	Beijing, China Sep, 2009 – Jul, 2012	Huazhong University of Science and Technology • B.E. in Automation	Wuhan, China Sep, 2005 – Jul, 2009
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RESEARCH INTERESTS	Creative AI, Generative Model, Computer Vision, Machine Learning						
RESEARCH OUTPUT STATISTICS	• Citations - Google Scholar: 15178 (H-index: 57) • 90+ papers at the best conferences and journals in computer vision and machine learning, including IEEE TPAMI (8), IJCV (6), Nature Communications (1), IEEE TIP (9), CVPR (23), ICCV (8), ECCV (11), NeurIPS (5), ICML (4), ICLR (6), SIGGRAPH (2), AAAI (3), ACL (2), ACM MM (3), Artificial Intelligence (1), IEEE TVCG (1), IEEE TKDE (2) • 10+ papers are recognized as Highly Cited Papers or Hot Papers by Clarivate						
AWARDS AND HONORS	• IEEE Signal Processing Society Outstanding Editorial Board Member Award, 2026. • AAAI Outstanding Senior Program Committee Service Award, 2026 • CCF-Tencent Rhino-Bird Faculty Fund Excellence Award, 2024 • Champion award in the Bracketing Image Restoration and Enhancement Challenge - Track 1 & 2, NTIRE 2024, in conjunction with CVPR 2024 • The second place in the Few-shot RAW Image Denoising competition of MIPI 2024, in conjunction with CVPR 2024 • CCF-Tencent Rhino-Bird Fellowship, 2023. • Top 2% Scientists Worldwide, 2023, 2024 & 2025. • ACM China Rising Star (Guangzhou Chapter), 2022. • CVPR Best Paper Finalist, 2019.						
RESEARCH GRANTS	Grants as Principal Investigator • Research on Generalizable Visual Restoration for Open-World Scenarios, AIS Support Fund for Interdisciplinary Research Collaboration, 2026/09 - 2027/08						

- Towards Efficient, Adaptive, and Scalable Visual Generative Models for Edge Devices, General Research Fund, Research Grants Council of Hong Kong, 2026/07 - 2029/06
- Large Model for Multimodal Understanding, Gift Fund, Baidu, 2026/07 - 2027/07
- AIGC video generation/editing and world models in film and television, Gift Fund, Tencent, 2026/07 - 2027/06
- Controllable High-Precision 3D Scene Generation Based on Spatial Relationship Awareness and Multimodal Guidance, Gift Fund, Tencent, 2025/12 - 2026/12
- Efficient Training Methods for Image Generation Models, Gift Fund, Kuaishou, 2025/12 - 2026/12
- Coherent Multi-subject Video Generation, Contract Research, ByteDance, 2025/09 - 2026/09
- Collaborative Project on Chain-of-Thought Reasoning Technology for Visual Generation, Contract Research, Huawei, 2025/07 - 2026/07
- AI Digital Human Research, Gift Fund, Video Rebirth Limited, 2025/07 - 2029/07
- New Architecture of Large Model, Gift Fund, Hong Kong Wiener Intelligence Technologies Limited, 2025/01 - 2026/12
- Research on Large-scale Fine-tuning and Alignment of Mixture-of-expert Large Models, HKUST-MetaX Joint Lab Fund, 2024/11 - 2026/09
- Content Restoration and Generation, Gift Fund, Huawei, 2024/09 - 2027/09
- Research on Key Algorithms of Face Restoration, Perception and Understanding for Under-display Cameras, General Program, National Natural Science Foundation of China, 2024/01 - 2027/12
- Efficient Video Restoration, Contract Research, Samsung, 2023/07 - 2024/07
- Towards Cross-domain Matching: Research on Multi-path Model for Advertisement Relevance, Contract Research, Tencent, 2023/06 - 2024/09
- Research on Image and Video Restoration in Autonomous Driving Scenarios under Adverse Weather, General Program, Guangdong Basic and Applied Basic Research Foundation, 2023/01 - 2025/12

Grants as Co-PI

- Solid-Liquid Biphasic Spatial Domain Ultrasonic Ultra-Low Water Consumption Plant-Based Fiber Mechanical Extraction Mechanism - AgriFiber Pro: From Agricultural Waste to Product, Platform Research Programme, Hong Kong Innovation and Technology Commission, 2025/12 - 2027/12
- Research and Development of Key Technologies for Multimodal Recognition and Generative Artificial Intelligence under Large Models, as well as Development and Industrialization of Intelligent Sports Scenarios, International Cooperation Program, Qingdao Municipal Bureau of Science & Technology, 2025/04 - 2027/04
- Research and Development of Key Technologies for Intelligent Generation of High-Quality Video Covers of Massive Video, Technology Development Key Project, Shenzhen Science and Technology Innovation Commission, 2022/11 - 2025/10

PUBLICATIONS

1. H. Li, J. Wang, S. Cai, H. Sheng, B. Deng, J. Ye, **W. Luo**, EchoStyle: Unlocking High-Fidelity Video Stylization with Reverse Data Synthesis, *European Conference on Computer Vision (ECCV)*, 2026.
2. H. Sun, M. Yang, S. Yang, Z. He, X. Han, H. Fu, C. Guo, **W. Luo**, ROAR-3D: Routing Arbitrary Views for High-Fidelity 3D Generation, *European Conference on Computer Vision (ECCV)*, 2026.
3. P. Li, R. Khirrodar, J. Li, Y. Dong, C. Cao, Y. Liu, **W. Luo**, Y. Guo, S. Saito, LUNA: Learning Universal 3D Human Animation Beyond Skinning, *European Conference on Computer Vision (ECCV)*, 2026.
4. Y. Jiang, T. Wang, Z. Wang, C. Cao, J. Wu, **W. Luo**, W. Hu, J. Gao, C. Guo, *European Conference on Computer Vision (ECCV)*, 2026.

5. J. Deng, T. Weng, T. Yang, **W. Luo**, Z. Li, W. Jiang, LlamaSeg: Image Segmentation via Autoregressive Mask Generation, *European Conference on Computer Vision (ECCV)*, 2026.
6. Y. Luo, Y. Wang, X. Zou, S. Tang, X. Luo, Z. Zhou, L. Xiong, Y. Peng, C. Yang, N. Wang, H. Song, G. Zou, J. Shi, X. Zi, M. Gao, N. Jia, P. Yang, F. Yang, Z. Zheng, P. Wu, W. Dong, P. Dong, S. Guo, H. Han, S. Zhu, J. Chen, J. Luo, W. Zhai, Y. Xu, J. Chen, Y. Fan, L. Qu, X. Chen, J. Zhuang, H. Chen, C. Yu, X. Zhang, Q. Liu, F. Zhou, S. Zhang, **W. Luo**, X. Yao, Z. Zhang, *Nature Communications*, 2026.
7. G. Liu, B. Yang, Y. Zhi, Z. Zhong, L. Ke, D. Deng, H. Gao, Y. Huang, K. Zhang, H. Fu, **W. Luo**, Beyond VLM-Based Rewards: Diffusion-Native Latent Reward Modeling, *International Conference on Machine Learning (ICML)*, 2026.
8. Z. Ye, Q. Liu, C. Wei, Y. Zhang, X. Wang, P. Wan, K. Gai, **W. Luo**, Visual-Aware CoT: Achieving High-Fidelity Visual Consistency in Unified Models, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
9. F. Yang, X. Dong, X. Yu, **W. Luo**, W. Liu, K. Zhang, MRD: Multi-resolution Retrieval-Detection Fusion for High-Resolution Image Understanding, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
10. M. Li, D. Wang, Z. Sun, J. Zhang, **W. Luo**, J. Guo, STiTch: Semantic Transition and Transportation in Collaboration for Training-Free Zero-Shot Composed Image Retrieval, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
11. Y. Li, L. Liu, X. ZHANG, W. Xue, **W. Luo**, Y. Guo, Q. Tian, CogniEdit: Dense Gradient Flow Optimization for Fine-Grained Image Editing, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
12. M. Li, P. Li, Z. Zhang, J. Lu, C. Zhao, W. Xue, Q. Liu, S. Peng, W. ZHANG, **W. Luo**, Y. Liu, Y. Guo, UniSH: Unifying Scene and Human Reconstruction in a Feed-Forward Pass, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
13. L. Ke, H. Yin, G. Liu, Z. Lv, J. Guo, C. Li, **W. Luo**, Y. Yang, J. Lyu, FlowSteer: Guiding Few-Step Image Synthesis with Authentic Trajectories, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
14. Z. Ye, X. He, Q. Liu, Q. Wang, X. Wang, P. Wan, D. Zhang, K. Gai, Q. Chen, **W. Luo**, UNIC: Unified In-Context Video Editing, *International Conference on Learning Representations (ICLR)*, 2026.
15. Y. Li, Z. Wang, G. Li, W. Xue, **W. Luo**, Y. Guo, Pixel-Perfect Puppetry: Precision-Guided Enhancement for Face Image and Video Editing, *International Conference on Learning Representations (ICLR)*, 2026.
16. X. Liu, Y. He, L. Guo, X. Li, B. Jin, Y. Li, C. Chan, W. Xue, **W. Luo**, Q. Liu, Y. Guo, HiPrompt: Tuning-free Higher-Resolution Generation with Hierarchical MLLM Prompts, *International Journal of Computer Vision (IJCV)*, vol. 134, 2026.
17. Z. Kong, F. Gao, Y. Zhang, Z. Kang, X. Wei, X. Cai, G. Chen, **W. Luo**, Let Them Talk: Audio-Driven Multi-Person Conversational Video Generation, *Neural Information Processing Systems (NeurIPS)*, 2025.
18. Y. Cai, Z. Jiang, Y. Liu, C. Jiang, W. Xue, Y. Guo, **W. Luo**, Foundation Cures Personalization: Improving Personalized Models' Prompt Consistency via Hidden Foundation Knowledge, *Neural Information Processing Systems (NeurIPS)*, 2025.
19. Z. He, M. Yang, S. Yang, Y. Tang, T. Wang, K. Zhang, G. Chen, Y. Liu, J. Jiang, C. Guo, **W. Luo**, MaterialMVP: Illumination-Invariant Material Generation via Multi-view PBR Diffusion, *Proc. of International Conference on Computer Vision (ICCV)*, 2025. **(Highlight)**
20. T. Wang, P. Xia, B. Li, P. Jiang, Z. Kong, K. Zhang, T. Lu, **W. Luo**, MOERL: When Mixture-of-Experts Meet Reinforcement Learning for Adverse Weather Image Restoration, *Proc. of International Conference on Computer Vision (ICCV)*, 2025.
21. R. Guan, T. Liu, W. Tu, C. Tang, **W. Luo**, X. Liu, Sampling Enhanced Contrastive

- Multi-View Remote Sensing Data Clustering with Long-Short Range Information Mining, *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, vol. 37, pp. 5598-5612, 2025.
22. M. Li, X. Long, Y. Liang, W. Li, Y. Liu, P. Li, **W. Luo**, W. Wang, Y. Guo, Multi-View Large Reconstruction Model via Geometry-Aware Positional Encoding and Attention, *IEEE Transactions on Visualization and Computer Graphics (TVCG)*, vol. 31, pp. 8564-8577, 2025.
 23. X. Chen, Y. Li, H. Shi, B. Hu, **W. Luo**, Y. Wang, M. Zhang, VideoVista-CulturalLingo: 360° Horizons-Bridging Cultures, Languages, and Domains in Video Comprehension, *The 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.
 24. Z. Kong, L. Li, Y. Zhang, F. Gao, S. Yang, T. Wang, K. Zhang, Z. Kang, X. Wei, G. Chen, **W. Luo**, DAM-VSR: Disentanglement of Appearance and Motion for Video Super-Resolution, *ACM SIGGRAPH*, 2025.
 25. P. Li, S. Ma, J. Chen, Y. Liu, C. Zhang, W. Xue, **W. Luo**, A. Sheffer, W. Wang, Y. Guo, CMD: Controllable Multiview Diffusion for 3D Editing and Progressive Generation, *ACM SIGGRAPH*, 2025.
 26. Z. Jin, Y. Qiu, K. Zhang, H. Li, **W. Luo**, MB-TaylorFormer V2: Improved Multi-branch Linear Transformer Expanded by Taylor Formula for Image Restoration, *IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 47, pp. 5990-6005, 2025.
 27. T. Wang, K. Zhang, Y. Zhang, **W. Luo**, B. Stenger, T. Lu, T. Kim, W. Liu, H. Li, LLDiffusion: Learning Degradation Representations in Diffusion Models for Low-Light Image Enhancement, *Pattern Recognition (PR)*, vol. 166, pp. 111628, 2025.
 28. Z. Ye, H. Huang, X. Wang, P. Wan, D. Zhang, **W. Luo**, StyleMaster: Stylize Your Video with Artistic Generation and Translation, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025.
 29. P. Li, W. Zheng, Y. Liu, T. Yu, Y. Li, X. Qi, X. Chi, S. Xia, Y. Cao, W. Xue, **W. Luo**, Y. Guo, PSHuman: Photorealistic Single-image 3D Human Reconstruction using Cross-Scale Multiview Diffusion and Explicit Remeshing, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025.
 30. X. Mao, Z. Jiang, F. Wang, J. Zhang, H. Chen, M. Chi, Y. Wang, **W. Luo**, OSV: One Step is Enough for High-Quality Image to Video Generation, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025.
 31. J. Xue, H. Wang, Q. Tian, Y. Ma, A. Wang, Z. Zhao, S. Min, W. Zhao, K. Zhang, H. Shum, W. Liu, M. Liu, **W. Luo**, Towards Multiple Character Image Animation Through Enhancing Implicit Decoupling, *International Conference on Learning Representations (ICLR)*, 2025.
 32. X. Qi, Y. Wang, H. Zhang, J. Pan, W. Xue, S. Zhang, **W. Luo**, Q. Liu, Y. Guo, Co³Gesture: Towards Coherent Concurrent Co-speech 3D Gesture Generation with Interactive Diffusion, *International Conference on Learning Representations (ICLR)*, 2025.
 33. Y. Li, S. Jiang, B. Hu, L. Wang, W. Zhong, **W. Luo**, L. Ma, M. Zhang, Uni-MoE: Scaling Unified Multimodal LLMs with Mixture of Experts, *IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI)*, vol. 47, pp. 3424-3439, 2025.
 34. P. Li, Y. Liu, X. Long, F. Zhang, C. Lin, M. Li, X. Qi, S. Zhang, **W. Luo**, P. Tan, W. Wang, Q. Liu, Y. Guo, Era3D: High-Resolution Multiview Diffusion using Efficient Row-wise Attention, *Neural Information Processing Systems (NeurIPS)*, 2024.
 35. L. Li, P. Dong, Z. Tang, X. Liu, Q. Wang, **W. Luo**, W. Xue, Q. Liu, X. Chu, Y. Guo, Discovering Sparsity Allocation for Layer-wise Pruning of Large Language Models, *Neural Information Processing Systems (NeurIPS)*, 2024.
 36. R. Li, J. Yu, C. Li, **W. Luo**, Y. Yuan, G. Wang, DREAM: Domain-agnostic Reverse

- Engineering Attributes of Black-box Model, *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, vol. 36, pp. 8009-8022, 2024.
37. T. Zhou, Q. Ye, **W. Luo**, H. Ran, Z. Shi, J. Chen, APPTracker+: Displacement Uncertainty for Occlusion Handling in Low-Frame-Rate Multiple Object Tracking, *International Journal of Computer Vision (IJCV)*, vol. 133, pp. 2044–2069, 2025.
 38. Z. Kong, W. Zhang, T. Wang, K. Zhang, Y. Li, X. Tang, **W. Luo**, Dual Teacher Knowledge Distillation with Domain Alignment for Face Anti-spoofing, *IEEE Trans. on Circuits and Systems for Video Technology (TCSVT)*, vol. 34, pp. 13177-13189, 2024.
 39. J. Tan, H. Park, Y. Zhang, T. Wang, K. Zhang, X. Kong, P. Dai, Z. Liu, **W. Luo**, Blind Face Video Restoration with Temporal Consistent Generative Prior and Degradation-Aware Prompt, *The 32nd ACM International Conference on Multimedia (ACM MM)*, 2024.
 40. Z. Kong, Y. Zhang, T. Yang, T. Wang, K. Zhang, B. Wu, G. Chen, W. Liu, **W. Luo**, OMG: Occlusion-friendly Personalized Multi-concept Generation In Diffusion Models, *European Conference on Computer Vision (ECCV)*, 2024.
 41. B. Tang, K. Zhang, **W. Luo**, W. Liu, H. Li, Prompting Future Driven Diffusion Model for Hand Motion Prediction, *European Conference on Computer Vision (ECCV)*, 2024.
 42. L. Li, H. Sun, S. Li, P. Dong, **W. Luo**, W. Xue, Q. Liu, Y. Guo, Auto-GAS: Automated Proxy Discovery for Training-free Generative Architecture Search, *European Conference on Computer Vision (ECCV)*, 2024.
 43. L. Li, Z. Wei, P. Dong, **W. Luo**, W. Xue, Q. Liu, Y. Guo, AttnZero: Efficient Attention Discovery for Vision Transformers, *European Conference on Computer Vision (ECCV)*, 2024.
 44. L. Li, Y. Bao, P. Dong, C. Yang, A. Li, **W. Luo**, Q. Liu, W. Xue, Y. Guo, DetKDS: Knowledge Distillation Search for Object Detectors, *International Conference on Machine Learning (ICML)*, 2024.
 45. J. Huang, Z. Wang, T. Liu, **W. Luo**, Z. Chen, W. Zhao, M. Wang, DeMPAA: Deployable Multi-Mini-Patch Adversarial Attack for Remote Sensing Image Classification, *IEEE Trans. on Geoscience and Remote Sensing*, vol. 62, pp. 1-13, 2024.
 46. X. Chen, J. Tan, T. Wang, K. Zhang, **W. Luo**, X. Cao, Towards Real-World Blind Face Restoration with Generative Diffusion Prior, *IEEE Trans. on Circuits and Systems for Video Technology (TCSVT)*, vol. 34, pp. 8494-8508, 2024.
 47. T. Wang, K. Zhang, Z. Shao, **W. Luo**, B. Stenger, T. Lu, T. Kim, W. Liu, H. Li, GridFormer: Residual Dense Transformer with Grid Structure for Image Restoration in Adverse Weather Conditions, *International Journal of Computer Vision (IJCV)*, vol. 132, pp. 4541–4563, 2024.
 48. X. Qi, J. Pan, P. Li, R. Yuan, X. Chi, M. Li, **W. Luo**, W. Xue, S. Zhang, Q. Liu, Y. Guo, Weakly-Supervised Emotion Transition Learning for Diverse 3D Co-speech Gesture Generation, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.
 49. Y. Shao, S. He, Q. Ye, Y. Feng, **W. Luo**, J. Chen, Context-Aware Integration of Language and Visual References for Natural Language Tracking, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2024.
 50. Y. Li, B. Hu, **W. Luo**, L. Ma, Y. Ding, M. Zhang, A Multimodal In-Context Tuning Approach for E-Commerce Product Description Generation, *International Conference on Computational Linguistics (COLING)*, 2024.
 51. Y. Gao, W. Zhang, **W. Luo**, L. Ma, J. Yu, G. Xia, J. Ma, Aux-NAS: Exploiting Auxiliary Labels with Negligibly Extra Inference Cost, *International Conference on Learning Representations (ICLR)*, 2024.
 52. W. Wang, C. Du, T. Wang, K. Zhang, **W. Luo**, L. Ma, W. Liu, X. Cao, Punctuation-level Attack: Single-shot and Single Punctuation Can Fool Text Models, *Neural Information Processing Systems (NeurIPS)*, 2023.

53. Y. Yu, P. Zhang, K. Zhang, **W. Luo**, C. Li, Y. Yuan, G. Wang, Multi-Prior Learning via Neural Architecture Search for Blind Face Restoration, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 36, pp. 3057-3070, 2025.
54. J. Tan, X. Chen, T. Wang, K. Zhang, **W. Luo**, X. Cao, Blind Face Restoration for Under-Display Camera via Dictionary Guided Transformer, *IEEE Trans. on Circuits and Systems for Video Technology (TCSVT)*, vol. 34, pp. 4914-4927, 2024.
55. K. Zhang, T. Wang, **W. Luo**, W. Ren, B. Stenger, W. Liu, H. Li, M. Yang, MC-Blur: A Comprehensive Benchmark for Image Deblurring, *IEEE Trans. on Circuits and Systems for Video Technology (TCSVT)*, vol. 34, pp. 3755-3767, 2024.
56. T. Wang, G. Tao, W. Lu, K. Zhang, **W. Luo**, X. Zhang, T. Lu, Restoring Vision in Hazy Weather with Hierarchical Contrastive Learning, *Pattern Recognition*, vol. 145, pp. 109956, 2024.
57. T. Gao, Y. Wen, K. Zhang, J. Zhang, T. Chen, L. Liu, **W. Luo**, Frequency-oriented Efficient Transformer for All-in-one Weather-degraded Image Restoration, *IEEE Trans. on Circuits and Systems for Video Technology (TCSVT)*, vol. 34, pp. 1886-1899, 2024.
58. T. Zhou, Q. Ye, **W. Luo**, K. Zhang, Z. Shi, J. Chen, FnF Attack: Adversarial Attack against Multiple Object Trackers by Inducing False Negatives and False Positives, *Proc. of International Conference on Computer Vision (ICCV)*, 2023.
59. P. Cheng, L. Lin, J. Lyu, Y. Huang, **W. Luo**, X. Tang, PRIOR: Prototype Representation Joint Learning from Medical Images and Reports, *Proc. of International Conference on Computer Vision (ICCV)*, 2023.
60. Y. Qiu, K. Zhang, C. Wang, **W. Luo**, H. Li, Z. Jin, MB-TaylorFormer: Multi-branch Efficient Transformer Expanded by Taylor Formula for Image Dehazing, *Proc. of International Conference on Computer Vision (ICCV)*, 2023.
61. S. Wang, C. Nguyen, J. Liu, K. Zhang, **W. Luo**, Y. Zhang, S. Muthu, F. Maken, H. Li, Homography Guided Temporal Fusion for Road Line and Marking Segmentation, *Proc. of International Conference on Computer Vision (ICCV)*, 2023.
62. Y. Shao, Q. Ye, **W. Luo**, K. Zhang, J. Chen, InterTracker: Discovering and Tracking General Objects Interacting with Hands in the Wild, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023.
63. Z. Song, Z. Zhang, K. Zhang, **W. Luo**, Z. Fan, W. Ren and J. Lu, Robust Single Image Reflection Removal Against Adversarial Attacks, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2023.
64. Z. Kong, W. Zhang, F. Liu, **W. Luo**, H. Liu, L. Shen and R. Ramachandra, Taming Self-Supervised Learning for Presentation Attack Detection: De-Folding and De-Mixing, *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*, vol. 35, pp. 10639-10650, 2024.
65. T. Wang, K. Zhang, T. Shen, **W. Luo**, B. Stenger, T. Lu, Ultra-High-Definition Low-Light Image Enhancement: A Benchmark and Transformer-Based Method, *Proc. of the Association for the Advancement of Artificial Intelligence (AAAI)*, 2023. (**Oral Presentation**)
66. H. Fang, P. Xiong, L. Xu, **W. Luo**, Transferring Image-CLIP to Video-Text Retrieval via Temporal Relations, *IEEE Transactions on Multimedia (TMM)*, vol. 25, pp. 7772-7785, 2023.
67. L. Zheng, Y. Li, K. Zhang, **W. Luo**, T-Net: Deep Stacked Scale-Iteration Network for Image Dehazing, *IEEE Transactions on Multimedia (TMM)*, vol. 25, pp. 6794-6807, 2023.
68. Z. You, K. Yang, **W. Luo**, X. Lu, L. Cui, X. Le, Few-shot Object Counting with Similarity-Aware Feature Enhancement, *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023.
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73. Y. Wang, Guo Pu, **W. Luo**, Y. Wang, P. Xiong, H. Kang, Z. Lian, Aesthetic Text Logo Synthesis via Content-aware Layout Inferring, *Proc. of IEEE Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022.
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76. W. Niu, K. Zhang, **W. Luo**, Y. Zhong, Blind Motion Deblurring Super-Resolution: When Dynamic Spatio-Temporal Learning Meets Static Image. *IEEE Trans. on Image Processing (TIP)*, vol. 30, pp. 7101-7111, 2021.
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COURSES

- AMCC5210, Practice and Techniques for Visual Creatives, 2025/26 Fall, 2026/27 Fall (PG)
- ARIN 5201, Machine Learning for Computer Vision, 2025/26 Fall, 2026/27 Fall (PG)
- MAIE 5421, Computer Vision, 2025/26 Fall, 2026/27 Fall (PG)
- ARIN 5302, Medical Image Analysis, 2025/26 Spring (PG)
- EMIA4110, Practical Machine Learning, 2024-25 Fall, 2024/25 Spring, 2025/26 Spring (UG)
- EMIA4991, Interdisciplinary Capstone Project, 2024/25 Spring (UG)

	• EMIA6950C/F, Independent Study, 2024/25 Spring, 2025/26 Fall, 2025/26 Spring (PG) • AMCC6900C, Independent Study in Arts and Machine Creativity, 2025/26 Spring (PG) • EMIA4900B, Independent Study, 2025/26 Spring, 2025/26 Summer (UG) • EMIA2010B, Cross-disciplinary Seminar: Digital Media and Creative Arts, 2024/25 Fall (UG)
PROFESSIONAL SERVICE	• Senior Area Editor: IEEE Signal Processing Letters (2025 - present) • Associate Editor: IEEE Transactions on Image Processing (2025 - present), Neurocomputing (2024 - present), IET Computer Vision (2024 - present) • Guest Editor: Pattern Recognition (Jun 2026 -), Computer Vision and Image Understanding (Sep 2020 - Sep 2021) • Senior Area Chair: NeurIPS 2026 • Area Chair: ICLR (2027, 2026), NeurIPS 2025, ACM MM (2026, 2025), ICML (2026, 2025), IJCAI 2025, IJCNN 2025, BMVC (2025, 2024) • Conference Organization/Service: SIGGRAPH Asia 2025 (Conflict-of-Interest Coordinator), CVM 2025 (Workshop Chair), AsiaGraphics (Vice Secretary, 2025 - present) • Senior Program Committee: AAAI (2027, 2026, 2025, 2024, 2022), IJCAI (2026, 2024, 2023, 2021) • Workshop Organizer: Consistency in Video Generative Models: from Clip to Wild, in conjunction with AAAI 2026 • Journal Reviewer: TPAMI, TIP, CSUR, TNNLS, TMM, TCSVT, PR • Conference Reviewer: ICML, NeurIPS, CVPR, ACL, ECCV, ICCV, BMVC, AAAI, IJCAI, ACM MM, WACV
UNIVERSITY AND SCHOOL SERVICES	• Interdisciplinary PG Studies Committee, Academy of Interdisciplinary Studies, 2025 - present • Search Committee, for the Head of the Div. Arts and Machine Creativity, 2024
DEPARTMENTAL SERVICES	• Chair of Undergraduate Program Committee, Div. of Arts and Machine Creativity, 2026 - present • PG Programs Coordinator, Div. of Arts and Machine Creativity, 2025 - present • Chair of PG Committee, Div. of Arts and Machine Creativity, 2025 - present • Search and Appointment Committee (SAC), Div. of Arts and Machine Creativity, 2024 - present • 3-in-1 Planning Committee (Research Matters), Div. of Arts and Machine Creativity, 2025 - present • Deputy Critical Incident Coordinator, Div. of Arts and Machine Creativity, 2025 - present • Search and Appointment Committee (SAC), Div. of Emerging Interdisciplinary Areas, 2024 - present • PG Committee, Div. of Emerging Interdisciplinary Areas, 2024 - 2025
TALKS	• Invited talk in the Young Scientists Forum in Lianhua Mountain Artificial Intelligence Conference 2026, Jul 2026 • Invited talk in The Forum on Low-Quality Visual Processing and Quality Assessment in Chinese Congress on Image and Graphics 2026, May 2026 • Invited talk in The Asiagraphics Workshop on Intelligent Graphics in SIGGRAPH Asia 2025, Dec 2025 • Invited talk in Tsinghua University, hosted by Prof. Yujiu Yang, Sep 2025

- Invited talk in the Institute of Automation, Chinese Academy of Sciences, hosted by Prof. Jin Gao, Aug 2025.
- Invited talk in The Forum on Multimodal Content Understanding and Generation in China Spatial Intelligence Conference 2025, Jul 2025
- Invited talk in the Institute of Automation, Chinese Academy of Sciences, hosted by Prof. Jin Gao, Aug 2025.
- Shenzhen University, hosted by Prof. Yanshan Li, Mar 2025.
- Shenzhen University, hosted by Prof. Xu Wang, Dec 2024.
- Zhejiang University, hosted by Prof. Qi Ye, Oct 2024.
- Southern University of Science and Technology, hosted by Prof. Feng Zheng, Jul 2024.
- Wuhan University, hosted by Prof. Guisong Xia, Apr 2024
- Zhejiang University, hosted by Dr. Qi Ye, Jun 2022.
- Beijing Institute of Technology, hosted by Prof. Changsheng Li, Jan 2022.
- Southern University of Science and Technology, hosted by Prof. Xiaoying Tang, Sep 2020.
- The Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, hosted by Prof. Yun Zhang, Sep 2018.
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PATENTS

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- **W. Luo**, L. Ma, W. Liu, Neural network model training method and device, and time-lapse photography video generating method and device, US 2020/0293833 A1
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- K. Zhang, **W. Luo**, L. Ma and W. Liu, Microexpression-based image recognition method and apparatus, and related device. US20210174072A1
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THESIS

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